Calculation Process:

1. Use the arrival cost as the performance measure.
2. Choose two algorithms that are similar (e.g., arrival, IS, etc.).
3. Trade a large enough number of orders in each algorithm in order to generate a representative sample size. Ensure that the orders traded in each algorithm have similar characteristics such as side, size, volatility, trade time, and market cap, and were traded in similar market conditions.
4. Let X = set of results from algorithm A.
   Let Y = set of results from algorithm B.
5. Categorize the data into groups of buckets.
   Combine the data into one series. Determine the bucket categories based on the combined data. We suggest using from ten to twenty categories based on number of total observations. The breakpoints for the category buckets can be determined based on the standard deviation of the combined data or based on a percentile ranking of the combined data. For example, if using the standard deviation method use categories such as $<-3\sigma$, -3σ to -2.5σ, ..., 2.5σ to 3σ, $3\sigma+$. If using the percentile ranking method order all data points from lowest to highest and compute the cumulative frequency from 1/n to 100% (where n is the combined number of data point). Select break points based on the values that would occur at 10%, 20%, ..., 100% if ten groups, or 5%, 10%, ..., 95%, 100% if twenty buckets. Count the number of data observations from each algorithm that fall into these bucket categories.
6. Compute the test statistic χ^2

$$\chi^2 = \sum_{k=1}^{m} \frac{(\text{observed sample X in bucket k} - \text{observed sample Y in bucket k})^2}{\text{observed sample Y in bucket k}}$$

   m = number of buckets.

Comparison to Critical Value:

- Reject the null hypothesis if $\chi^2 \geq \chi^{2*}(df = m-1,\ \alpha = 0.05)$

Kolmogorov-Smirnov Goodness of Fit

The Kolmogorov-Smirnov goodness of fit test is used to determine whether two data series of algorithmic performance could have been generated from the same underlying distributions. It is based on the cumulative distribution function (cdf). If it is determined that the data samples could not have been generated from the generating process then we conclude that the algorithms are different.